SIEMENS

Sensation

CT

Maintenance Instructions

System

Maintenance System

This document is valid for:
SOMATOM Sensation 10/16/Cardiac
SOMATOM Sensation 40/64/Cardiac 64
SOMATOM Sensation Open

The protocol CT02-023.832.01.19.02 is required for these instructions

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The document corresponds to the version/revision level effective at the time of system delivery. Revisions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order current revision levels.

Disclaimer

The installation and service of equipment described herein is to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services.

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1 General

1.1 System maintenance intervals

Partial maintenance	every 6 months
Full maintenance	every 12 months
Preventive parts replacement	every 3 years

1.2 System maintenance times

1.2.1 Partial maintenance

Partial maintenance, performed every **6 months**, consists of the following parts:

Preparations	20 min
Preventive maintenance - part 2	90 min
Image quality check - constancy	30 min
Final steps (without “checking the protective system conductor” step 8.2)	20 min

1.2.2 Full maintenance

Full maintenance, performed **every 12 months**, consists of the following parts:

Preparations	30 min
Safety inspections	60 min
Preventive maintenance - part 1	60 min
Preventive maintenance - part 2	90 min
Image quality check - constancy	30 min
Final steps	30 min

1.2.3 Preventive parts replacement

Preventive parts replacement, performed **every 3 years**, consists of the following parts:

Preventive parts replacement	30 min

1.3 System maintenance planning

1.3.1 Partial maintenance

The partial maintenance is performed every 6 months and grouped into 3 packages as follows:

- Safety inspections
- Preventive maintenance - part 2
- Image quality check - constancy

1.3.2 Full maintenance

The full maintenance is performed every 12 months and includes the partial maintenance work as well. To improve planning and maintenance, the individual tasks have been grouped into 4 packages as follows :

- Safety inspections
- Preventive maintenance - part 1
- Preventive maintenance - part 2
- Image quality check - constancy

Each of these 4 packages may be performed independently of the others. Complete instructions are provided in the respective chapters. This makes it possible to schedule the full maintenance over several days. The work time required will be extended according to the respective 'Next steps' sections.

1.3.3 Preventive parts replacement

The UPS has to be replaced every 3 years beginning with the date of installation. The date for the next replacement has to be documented in the Maintenance Protocol at the time of the first maintenance and then at every subsequent maintenance interval.

1.4 Safety information

The installation, service and maintenance of equipment described herein is to be performed by qualified personnel who are employed by Siemens or one of its affiliates, or who are otherwise authorized by Siemens or one of its affiliates to provide such services. Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

In this context, "qualified" signifies that the service engineer has been trained and has had practical experience in the required routines so that he/she is capable of performing service work on the system. "Authorized" indicates that the service engineer is recognized by the system owner as a qualified service engineer and is therefore authorized to perform installation, service and, maintenance work on the system.

⚠WARNING

[hz_serdoc_F13G01U12M02]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

➪ **Read and observe the “General safety notes” and the “Product-specific safety notes”.**

⚠WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

➪ **The guidelines contained in this document have to be strictly followed.**

1.4.1 Bloodborne pathogens

⚠CAUTION

[hz_serdoc_F13G07U01M01]

Handling parts of the system that may have come into contact with patients may lead to infection through blood-borne pathogens.

Infection caused by blood-borne pathogens!

➪ **Take appropriate precautions against exposure to bloodborne pathogens (e.g. wear gloves).**

1.4.2 High voltage

⚠WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when the system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- **Ensure via measurements that all voltages are switched off.**
- **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

1.4.3 Radiation protection

Observe the following protective measures when working on the CT unit with radiation "ON"

- use a radiation shield
- maintain a safe distance from the source
- use the minimum dose, smallest slice-thickness
- remove phantom (if possible)

⚠CAUTION

[hz_serdoc_F13G01U03M01]

Not following X-rays protective regulations may lead to radiation exposure for yourself and/or other persons.

Risk of radiation exposure!

Observe radiation protection regulations when X-rays are switched on, e.g. :

- **Never work inside the gantry room.**
- **Do not leave the system unattended.**
- **Make sure there is no one inside the gantry room.**
- **Make sure that no person can enter the gantry room unnoticed (e.g. lock doors if necessary) .**

⚠CAUTION

`[ hz_serdoc_F13G01U03M02 ]`

Certain tests and procedures use X-rays. Observe radiation protection regulations (CT basic course).

Risk of exposure to X-rays.

- **It is prohibited to perform service on the system, if for any reason the appropriate training or safety instructions regarding X-ray protection were not received during the CT training or through other qualified sources.**

1.4.4 Working in the gantry

Prior to starting work in the gantry:

1. Switch the system off; observe the power-off instructions and wait times as described above.
2. Wait until rotation has stopped.
3. Disable gantry rotation via service switch S301
4. Always use the rotation safety bolt to lock the rotation part of the gantry.
5. Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry. Use protective conductor cable 4665329 B1945

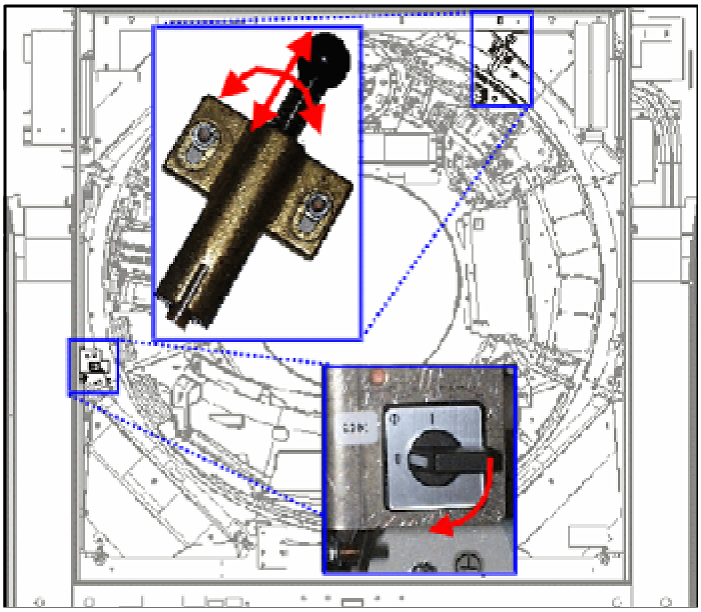


Fig. 1: Location of rotation safety devices

NOTE

Remove the cable and rotation safety bolt prior to rotating the gantry and enable gantry rotation (S301) after finishing work.

2 Prerequisites

2.1 Required tools and auxiliary materials

2.1.1 Documents

- Latest version of system **service documentation CT02-023.801.01** xx.xx
- **Maintenance Protocol** CT02-023.832.01. xx.xx **in printed form** .

2.1.2 Tools

- Standard tool kit
- Sliding calipers
- Torque wrench (3 Nm, 6 Nm, 28 Nm)
- Vacuum cleaner

2.1.3 Measurement devices and auxiliary materials

- Lubricant gun
- Tape measure
- Protective conductor meter (e.g. item number 44 158 99 RV090 or SECUTEST SIII))
- Service (termination) connector for gantry (back) doors

2.1.4 Test equipment

- Quality phantoms, delivered with the system

2.1.5 Consumable items (lubricant, filters)

Item	Item number	Amount	Required
Air filter for PDC	30 68 095	1	1
Air filter for dehumidifier	27 93 250	5	1

Item	Item number	Amount	Required
Air filter set for IRS Contains plastic frames and filters (small and large) for all 4 computers (IRS 1 and 2) or 4x 3 filter pads (IRS 3)	73 95 325 (for IRS 1) 73 95 473 (for IRS 2 / 2c / 2F) 83 79 344 (for IRS 3)	1	1
Air filter for IRSmx2s	101 61 765	1	1
Stator ring brush (not for Sensation 64/Cardiac 64 and Open)	73 95 101	1	1
Lubricant Isoflex Topas NCA 52	15 90 863	1 kg	n.a.

2.1.6 Spare parts

Item	Item number	Amount	Required
UDC Power brushes Set of 20 UDC power brushes including 2 caps	73 95 143	1	1
Low power carbon brushes* Set of 12 low power brushes, including 2 caps	73 95 176	1	1
Control data unit* includes 16 brushes	73 95 168	1	1

2.1.7 Cleaning materials

- Soft cloth for monitor screen
- Cleaning agent for system components (e.g. Plexiglas cleaner, mild soap , or commercially available cleanser)

2.2 Evaluating Reports

PM **Checking system reports**

NOTE	**If remote access to the system (RD) is available, this action should be performed prior to the maintenance call!**

To check for errors, the**| Reports |** function can be used to obtain a readout of the logbook:

1. Switch on the system, go to “Local Service” and start the service software.
2. Start | Reports | in the upper naviagation frame.
3. For additional information on how to handle the Eventlog, click the **| Help |** button in the upper navigation frame. Check the following reports:
   - **Tune up/QA/tests** --> **Tune up** --> **Table generations** --> **Defective channels**

     Click the results and check the “Table Editing History” and “Affected Modules” if new defective channels were detected by the system and added. Refer to “Trouble shooting guide Gantry” (DMS) and change ADM-Modules if necessary.
   - **Performance --> Statistics**

     Check results. If the system displays abormal behavior (e.g. a high number of “Restarts”), check the logbook for more details. The “Error Hitlist” may also be helpful for further investigation.
   - **Performance --> Error Hitlist**

     Check for errors that occur frequently. Follow the instructions in the description text for further actions.
   - **Tube --> Current History**

     Check the tube history for tube arcings. If necessary, getter the tube or continue with “Tube Errors”.
   - **Tube --> Errors**

     Check for errors that occur frequently. Follow the instructions in the description text for further actions.

2.3 Evaluating the logbook

PM Error statistics readout

NOTE	**If remote access to the system (RD) is available, this action should be performed prior to the maintenance call!**

Read out the logbook and check for errors:

1. Switch on the system; go to “Local Service” and start the service software.
2. Start the **I event log I** in the upper navigation frame.
3. Check the event log for errors. For additional information on how to handle the event log, click the **I Help I** button in the upper navigation frame. If necessary discuss the results with the customer
   - Depending on the results, discuss the system performance with the customer.
   - Determine what, if any, concerns the customer has with respect to the system and how they can be addressed.
   - Note any customer concerns and the actions taken to resolve them in the Comments section of the Maintenance Certificate.
4. A statisical evaluation of relevant errors can be perfomed using the **I Reports I** function.

3 Safety Inspections

3.1 General

⚠WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

- **The guidelines contained in this document have to be strictly followed.**

3.1.1 Preparations

NOTE

Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.

3.1.2 Required tools and auxiliary materials

Not required.

3.2 Function Tests

To perform the following function tests, the system has to be switched on and in working condition.

3.2.1 Measuring the patient leakage current (DIN VDE 751-1)

SI Patient leakage current checked

Note :

This measurement is required in Germany only, or in parts of German-speaking countries (e.g. Austria, Switzerland). Since no direct measurements are required for this system in accordance with DIN VDE 751-1, a justification is provided for the sake of completeness. For this reason, the justification is provided in German only:

System leakage current: not required.

Begründung: Festanschluß und bauseitige Schutzmaßnahmen bei indirektemBerühren (z.B. nach DIN VDE 0107)

patient leakage current at the part of the table being used (type B): not required

Begründung: Trennung zwischen spannungsführenden Teilen und Anwendungsteil durch schutzleiterverbundene Metallteile nach IEC 60601-1 , Abschnitt 17.a.2

patient leakage current at the ECG module (type BF) integrated into the table: not required

Begründung: Konstruktive Maßnahmen und Festeinbau

3.2.2 Checking the protective ground conductor resistance

This test checks the protective ground conductor resistance of the system.

NOTE	**During maintenance several of the protective ground conductors will be handled. As a result, the protective conductor resistance measurement will be performed in the last chapter of this document. Refer to (Checking the protective system conductor / p. 65) for more information.**

3.2.3 Checking the function of the door contact switch

SI Door contact switch checked

This test checks whether a door switch blocks radiation.

NOTE	**Omit this test if a door switch for the examination room has not been installed.**

- Select any scan mode with radiation. To minimize radiation, select a mode with low dose, small slice settings.
- Press **Start** for the measurement and open the scan room door while radiation is ON.
  ➪ The test is successful if the scan is blocked and a message appears.

3.2.4 System: Checking the radiation preparation indicator

SI Radiation preparation indicators checked

This test checks the function of the installed radiation preparation indicators.

NOTE	**Omit this test if a radiation preparation indicator (usually a yellow warning lamp) has not been installed.**

- Select any scan mode with radiation. To minimize radiation, select a mode with low dose, small slice settings.
- Load any mode and wait until the system prompts you to press the **Start** button. The radiation preparation indicators have to be ON.
- Press **Start** for the measurement.
- The indicator has to be ON during the scan.

3.2.5 System: Checking the radiation indicator

SI Radiation indicators checked

This test checks the function of the installed radiation “ON” indicators.

NOTE	**This test is intended to check the radiation indicatiors of the system. If additional (optional) radiation indicators (radiation warning lamps) were installed, they have to be checked accordingly.**

- As described above, the system is still in scan preparation. (Any scan mode with radiation was previously selected).
- Press **Start** for the measurement and check whether
  - the acoustic radiation indicator in the control box is audible.
  - all radiation warning lamps (gantry front and back, control box) are ON

  were ON while radiation is applied.

3.2.6 System: Checking the radiation monitor (108%)

SI Radiation switch-off checked

This test checks the function of the X-Ray radiation monitor.

NOTE	**The TIMEOUT TEST checks whether the radiation monitor in the generator is functioning properly. Load a scan mode with a higher number of readings than required for the respective scan time. After a radiation period of 110% (nominal scan time plus 10%), the generator must switch off radiation and transmit an error message.**

- Open the test platform the for X-ray - Timeout test under **Test Tool --> Controller --> XRS --> XRay - Timeout** .
- Start and evaluate one test for each mode (topo, sequence and spirals).
  - Topogram
  - Sequence
  - Spiral (select “min” and “max” for measurement)
    - ➪ The test is successful, if the timeout value is **< 110%** .

3.2.7 System: Checking the emergency STOP circuit

SI Emergency STOP circuit checked

This test checks the function of the emergency STOP circuit.

NOTE	**This test checks whether all motor-driven system movements are immediately stopped when the EMERGENCY STOP button is actuated.**

- Select any scan mode with radiation and rotation. To minimize radiation, select a mode with low dose, small slice settings.
- Press the red EMERGENCY STOP button on the keyboard while the gantry is rotating.
  - ➪ This should stop all motor-driven system movement (gantry tilt, gantry rotation, table movement)
  - ➪ The system is now in STOP status.
  - ➪ A window appears with the request to <continue>.
- Confirm the pop-up window to continue measurement.
- Reload the mode and repeat the above test procedure for the EMERGENCY STOP button on all gantry control panels (front and rear).

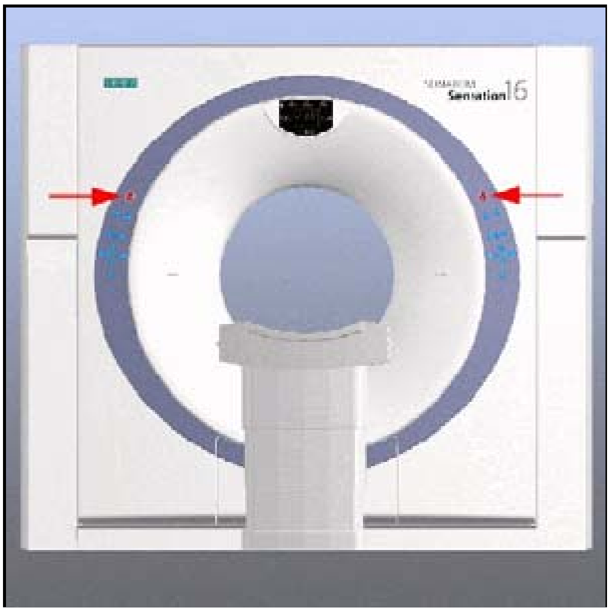


Fig. 2: Gantry front: location of E-Stop buttons

3.2.8 Gantry: Checking the tube cooling hoses

SI Tube cooling hoses checked

⚠CAUTION

[hz_serdoc_F13G02U01M01]

Temperature of X-ray tube and/or tube cooling device parts may be above 70°C (with STRATON up to 130°C).

Risk of burns!

- ➪ **Avoid contact with oil and exposed parts of X-ray tube and cooling device.**
- ➪ **Wait until X-ray tube and cooling device parts are chilled down.**

⚠CAUTION

[hz_serdoc_F04G05U01M04]

Injury from hot cooling oil may result, if cooling hoses become damaged.

Risk of injury!

- ➪ **Check the correct position of the cooling hoses and couplings as described in the maintenance instructions.**

- Check the hoses of the tube cooling system
  - ➪ They should not be loose or improperly routed
- Inspect all cooling hoses and cooling devices for oil leaks.

3.2.9 PHS: Checking the safety switches

SI Safety limit switches checked

This test checks the function of the PHS safety switches.

NOTE	**Safety switches stop horizontal table travel when the foot-end top support makes contact with an obstacle or when someone steps on the floor-mounted connecting frame between the gantry and the table.**

Activate the following safety switches:

- **PHS foot-end safety switch** by pressing the top support cover located at the foot-end toward the gantry
- **Floor-mounted connecting frame (PHS <--> Gantry) safety switch** by stepping on the black connecting frame on the floor between PHS and the gantry.

In each casek, you will hear a distinct click.The GPC disables all push buttons for horizontal movement on the gantry panels. The corresponding push buttons are dimmed.

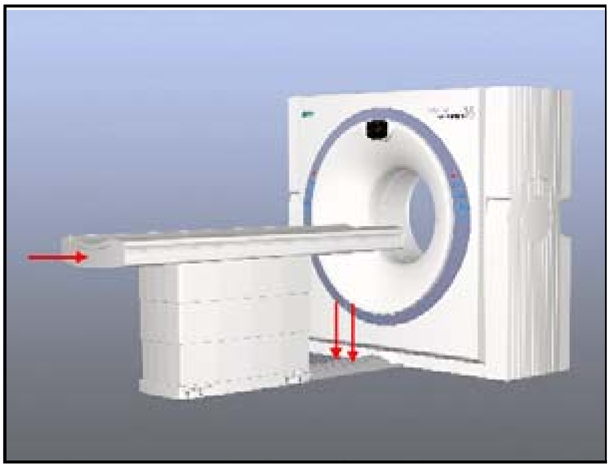


Fig. 3: PHS: location of safety switches

3.2.10 PHS: Checking the spindle brake

SI Function of spindle brake checked

This test checks the function of the PHS spindle brake.

- Enable the service functions by clicking: **Option --> Local Service** .
- Go to **Test Tool --> Controllers --> PTV**
- Select **Spindle brake** and press **<GO>** to start the test.
- Follow the test results on screen.

3.2.11 PHS: Checking the motor brake

SI Function of motor brake checked

This test checks the function of the PHS motor brake.

- Enable the service functions by clicking: **Option --> Local Service** .
- Go to **Test Tool --> Controllers --> PTV**
- Select **Motor brake** and press **<GO>** to start the test.
- Follow the test results on screen.

3.2.12 PHS: Checking the patient positioning aids

SI Patient positioning aids checked

NOTE	**This section tests the patient positioning supports attached to the patient table.**

- Inspect the patient positioning accessories for damage (i.e. head cradle, holder, pads).
- Check the accessory bracket at the head-end of the table for damage.
- Check the lock-in mechanism on the patient positioning accessories.
  - Attach them and remove them from the accessory bracket at the head-end of the table several times.
  - Test criteria: the accessories should be safely and securely positioned on the patient table.

NOTE	**If the locking mechanism does not function properly, determine the cause of the problem (i.e. spring, knob, or bracket insert in table) and replace the appropriate part(s).**

3.2.13 PDC: Protection functionality test

SI Circuit breaker checked

This test checks the function of the main circuit breaker F99.

1. Verify that F1 and F9 breakers in the PDC are switched on.

   ➪ If fuse F9 is switched off, the main breaker F99 will not work.
2. Switch on the power using the onsite ON/OFF switch. Make sure Somaris was shut down on the Navigator and Wizard.
3. Verify the status of the K100 and K101 LED"s (Tab. 1 / p. 23) .
4. If one of the LED"s has the wrong status, switch the power off and on using the onsite ON/OFF switch. If the problem persists, check all wiring

5. **Short Circuit Test**

   If the status of the LEDs is o.k., press the S100 test key (3/Fig. 4 / p. 23) and hold for at least 2 seconds.

   ➪ The main breaker F99 must trip off to the middle position.

6. To switch on the main breaker F99, first move the breaker to the OFF position and then to the ON position (requires extended pressure).
7. Verify the status of the K100 and K101 LEDs (Tab. 1 / p. 23) .
8. **Leakage current test**

   If the LED"s status is o.k., press the S101 test key (4/Fig. 4 / p. 23) and hold for at least 2 seconds.

   ➪ The main breaker F99 must trip off to the middle position.

9. To switch on the main breaker F99, first move the breaker to the OFF position and then to the ON position (requires extended pressure).

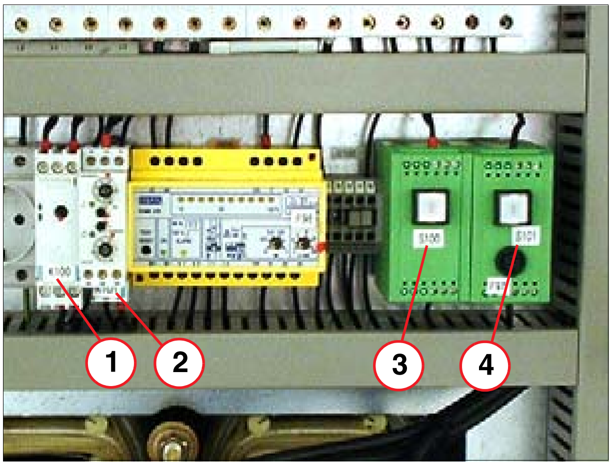


Fig. 4: K100 and K101

Pos. 1 K100

Pos. 2 K101

Pos. 3 S100

Pos. 4 S101

Tab. 1 LED"s K100/K101

LED	green (230 V)	yellow	yellow
K100 Time delay relay	n.a.	OFF	OFF
K101 Current control relay (DCC T1)	ON	ON	n.a.

3.3 Next steps

NOTE

Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained. Depending on the maintenance schedule, you may now conclude your maintenance activities with the concluding test or proceed with additional maintenance work according to the schedule (Recommended (Preventive maintenance (part 1) / p. 25) .

3.3.1 Labels

If any label has peeled off from the system or any component, replace the label. Refer to the section **Location of Labels** in the **System Owner Manual**, which must always be kept on hand and stored with the system, to identify the label and the correct location of the label.

3.3.2 Concluding test

Q **Condition of the CT system checked**

- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

4 Preventive maintenance (part 1)

4.1 General

⚠WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

➪ **The guidelines contained in this document have to be strictly followed.**

The work steps in this section may not include all of the necessary details. If you require more detailed instructions (e.g. on how to remove the table covers), please refer to the latest version of the service documentation “PHS - Replacement of parts” - CT02-023.841.03.xx.xx, for further details.

4.1.1 Preparations

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the ’Maintenance Protocol’ provided.

- Move the table to the lowest vertical position.
- Remove
  - the center mounting screw of the base cover (front side)
  - the center mounting screw of the base cover (back side)
- Move the table to maximum height.
- Switch off the system.

⚠WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when the system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- ➪ **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- ➪ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- ➪ **Ensure via measurements that all voltages are switched off.**
- ➪ **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

4.1.2 Required tools and auxiliary materials

- Spring scale
- Phantom holder, no. 73 96 943, delivered with the system
- Standard service kit
- Lubricant TOPAS NCA 52 (item number: 15 90 863)

4.2 Patient handling system

4.2.1 Lubricating the spindle nut

PM Spindle nut lubricated

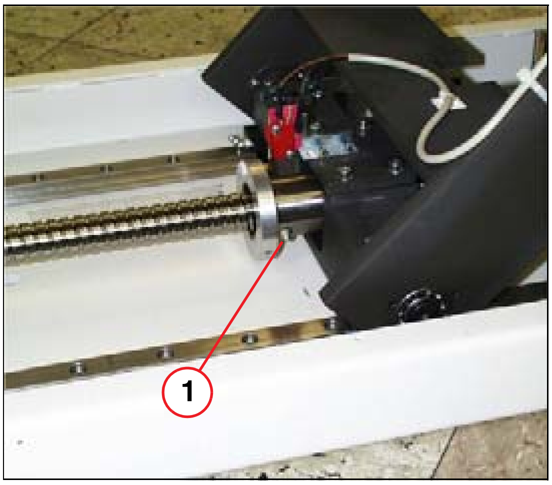


Fig. 5: Spindle: location of grease nippel
Pos. 1 Grease nippel

- Place the connector of the lubricant gun on the lubrication fitting of the spindle nut (1/Fig. 5 / p. 27).
- Pump lubricant into the spindle nut until lubricant flows out of both sides.
- Clean any old lubricant from the spindle.
- Remove the connector of the lubricant gun from the spindle nut.

4.2.2 Lubricating the linear guides on the lower part of scissor mechanism

PM Lower part of the scissor mechanism: Linear guides lubricated

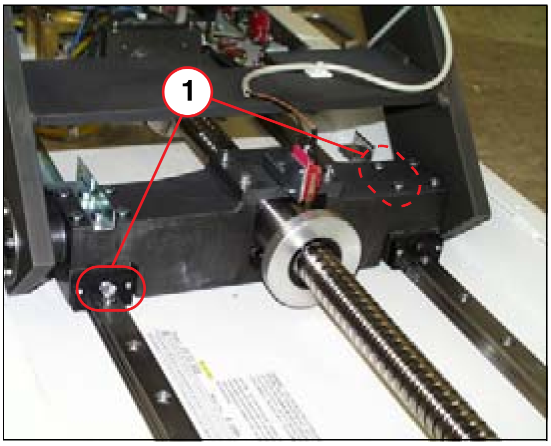


Fig. 6: Scissor: location of grease nippel
Pos. 1 Location of grease nippel

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/Fig. 6 / p. 27).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.3 Lubricating the linear guides of the upper part of scissor mechanism

PM Upper part of the scissor: Linear guides lubricated

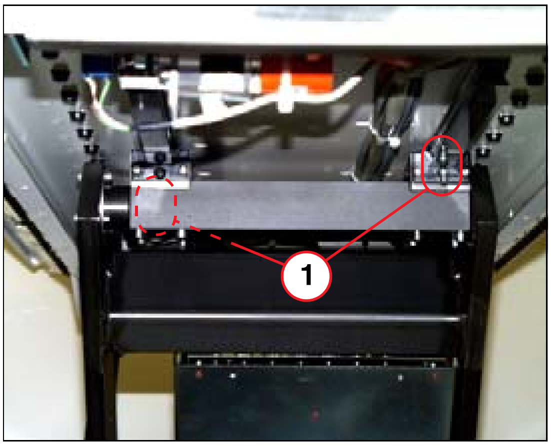


Fig. 7: Scissor: location of grease nippel
Pos. 1 Location of grease nippel

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/Fig. 7 / p. 28).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.4 Lubricating the guide rails of the tabletop

PM Tabletop: Guide rails lubricated

- To access the guide rails, remove the cover(s)
  - under the tabletop at the head-end of the table
  - at the foot-end of the table
- Move the tabletop accordingly to access the grease fittings shown.

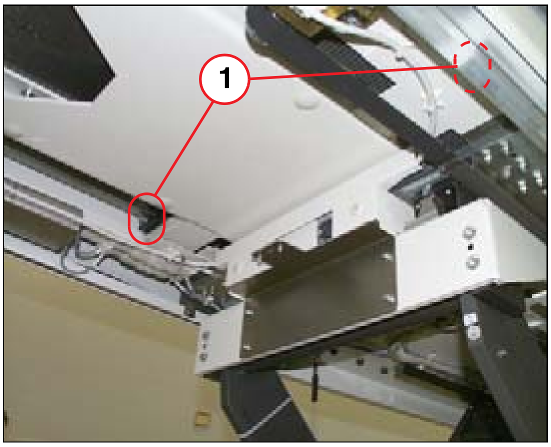


Fig. 8: Tabletop: location of grease fitting
Pos. 1 Grease fitting

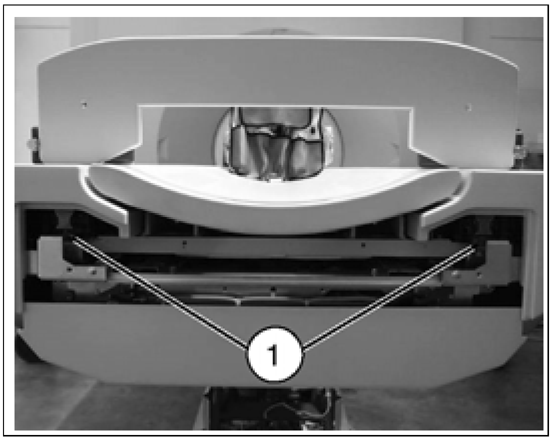


Fig. 9: Linear guides of the tabletop (foot-end of table)

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/Fig. 9 / p. 29).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.5 Lubricating the guide rails of the top support

PM **Top support: Guide rails lubricated**

- Move the tabletop accordingly to access the grease fittings shown.

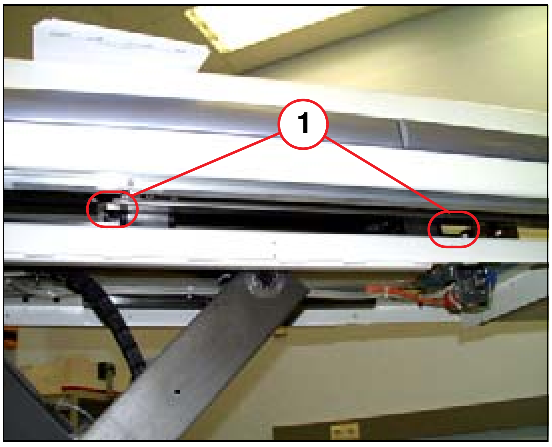


Fig. 10: Table: location of grease fitting
Pos. 1 Grease fitting

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/Fig. 10 / p. 29).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.6 Checking the movement of the tabletop

PM Movement force for horizontal tabletop travel checked

- Insert the phantom holder delivered with the system into the socket on the tabletop.
- Secure the phantom holder to the tabletop using a cable tie.
- Move the tabletop toward the gantry (approximately 100 mm).
- Move the top support of the table toward the gantry (approximately 100 mm).
- Hook the spring gauge into the cable tie.
- Using the spring gauge, pull the tabletop toward the foot-end of the table. Maintain a constant force.
  - View the spring gauge to check the force required to move the tabletop.

NOTE	**The force required to move the tabletop should be < 160 N.**

4.2.7 Checking the movement of the top support

PM Movement force for horizontal top support travel checked.

- Pull the tabletop toward the foot-end until it reaches the end stop. (The top support is still positioned about 100 mm toward the gantry)
- Using the spring gauge, pull the top support toward the foot-end of the table. Ensure that the force remains constant.
  - View the spring gauge to check the force required to move the tabletop.

NOTE	**The force required to move top support should be < 160 N.**

- Remove the cable tie, the phantom holder, and the spring scale.

4.2.8 Checking the compression spring on the patient table

PM Compression spring checked

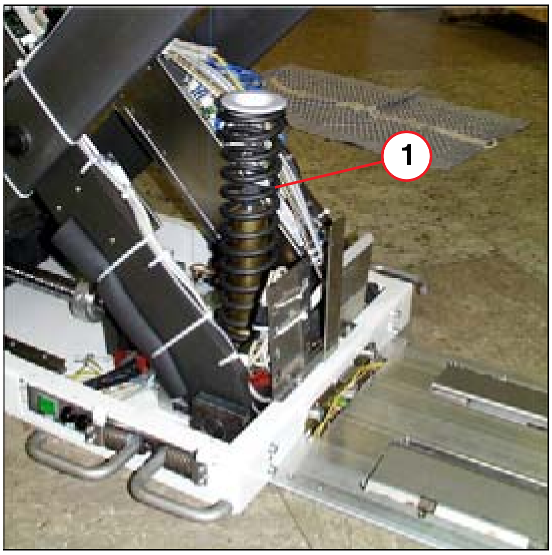


Fig. 11: Location of compression spring
Pos. 1 Compression spring

- Check the spring for mechanical damage such as cracks or breakage.

4.2.9 Cleaning the motor brake of the couch lift motor

PM Vertical motor brake cleaned

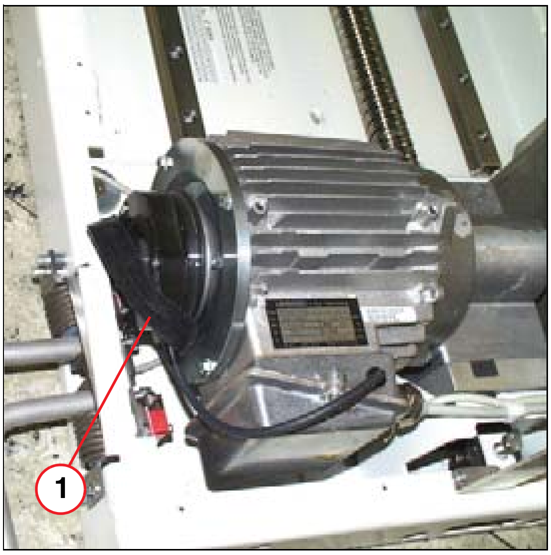


Fig. 12: PHS vertical motor
Pos. 1 Location of the rubber gasket

- Pull off the rubber gasket (1) from the vertical motor brake.
- Tap the vertical motor brake lightly or switch the drive on and off briefly to dislodge the abrasion dust from the motor brake.
- Clean the motor brake carefully and remove dust using a vacuum cleaner.
- Reinstall the rubber gasket (1) on the vertical motor brake

4.3 Next steps

- Reattach all covers on the PHS.

  If necessary, use the service documentation, “PHS- Replacement of parts” - CT02-023.841.03.xx.xx, for further details.

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the ’Maintenance Protocol’ provided. Depending on the maintenance schedule, you may either conclude the maintenance activities with the concluding test or proceed with additional maintenance work according to the schedule (Recommended (Preventive maintenance (part 2) / p. 33).

4.3.1 Concluding test

Q Condition of the CT system checked

- Switch on the system.
- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

5 Preventive maintenance (part 2)

5.1 General

⚠WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

➪ **The guidelines contained in this document have to be strictly followed.**

Perform the maintenance workflow according to this document.

The work steps in this section may not include all of the necessary details. If you require more detailed instructions (e.g. how to remove the gantry covers or the gantry tunnel), please refer to the latest version of the service documentation “Gantry - Replacement of parts” - CT02-023.841.01.xx.xx, for further details.

5.1.1 Preparations

- Switch on the system.
- Move the table to the lowest position.
- Open the front door of the gantry.
- Disable gantry rotation via service switch S301.

5.1.2 Required tools and auxiliary materials

- Standard service kit
- Sliding calipers
- Lubricant: Isoflex Topas NCA 52 (15 90 863)
- Grease gun
- Tape measure (only for the Care Vision option)

5.2 Care Vision (Option)

5.2.1 Checking the weight compensation of the support arm

SI Weight compensation of the support arm checked

NOTE	During this procedure, other maintenance activities can be performed.

- Move the support arm to a high position.
- The support arm must remain in this position for about 30 minutes.

  To check this, measure the distance between two reference points, e.g. at the monitor and the tabletop using a tape measure. Note the distance and repeat the measurement after 30 minutes.
- Repeat this procedure also in the center and a lower position of the support arm.
- In case the support arm is moving, it has to be adjusted according to(CT00-000.841.02 / Care Vision (beginning with SW version VB 20))
- If the weight compensation cannot be adjusted because the carriage is at the end stop, the gas spring has to be replaced.

5.3 Sliding gantry (optional)

PM Check the toothed belt

- The toothed belt has to be **checked annually**.
  - ➪ The adjustment for the torque of the toothed belt tension must be **1.8 +0.2 Nm for a roadway <=3.2 meter** and **2.5 +0.2 Nm for a roadway >3.2 meter** on each tensioning screw.

NOTE — **For adjustment, the gantry must be positioned fully toward the side of the drive motor.**

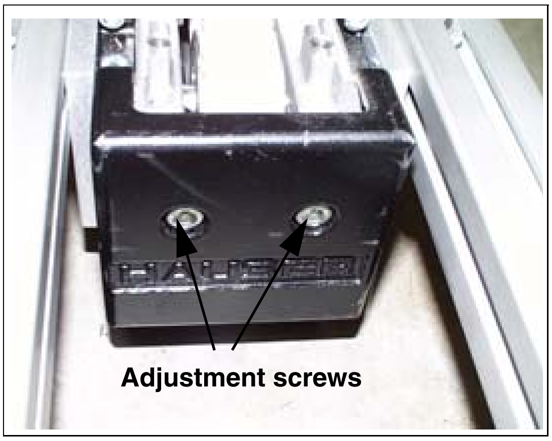


Fig. 13:

Fig. 14:

NOTE — **The toothed belt must run centered on the toothed wheel. Adjust it in small steps alternately with both tensioning screws. Check the centrical running by moving the gantry in both directions.**

- Check the rails for possible foreign particles
  - ➪ Open the metal covering as shown in the image and check the rail inside.

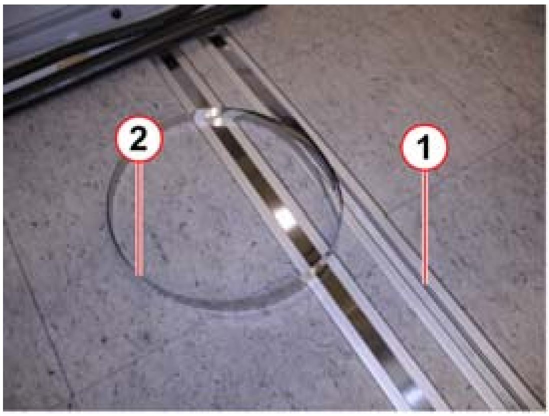


Fig. 15: Rails

Pos. 1 Rail

Pos. 2 Metal covering

Fig. 16: Checking the rail inside

PM Cleaning the energy chain (every 3 years)

- Clean the energy chain (every 3 years).
- Remove the lower covers of the energy chain and clean with a vacuum cleaner.

PM Check the switching elements (front and rear)

- Check the switching elements by moving the gantry in the forward and the reverse direction.
  - ➪ Push with your foot against the switching element to check the stop function.

PM Check the blind covering (optional)

- Check the blind covering for damages as well as the tension pulley for proper operation.
  - ➪ Move the sliding gantry twice over the entire roadway.

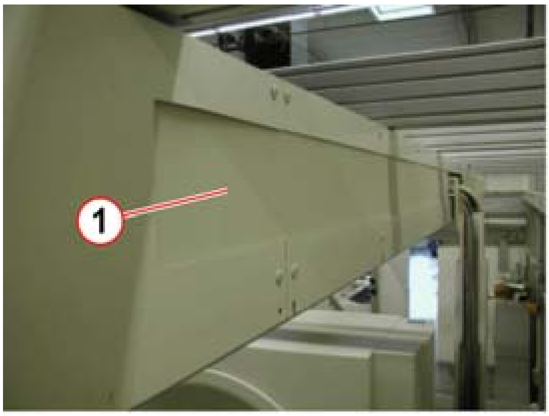


Fig. 17: Energy chain box

Pos. 1 Blind covering

PM **Check the protective conductor connection**

- Check both protective conductor cables between the power distribution cabinet (PDC) and the right stand of the gantry.
  - ➪ Disconnect both protective conductor cables in the PDC (Gantry PE301 and W2) for the measurement. (1/Fig. 18 / p. 37)(For checking the protective conductor cable for possible cable breaks in the energy chain).
  - ➪ Each protective conductor cable must be measured separately between the PDC and the gantry.
  - ➪ Reconnect the protective conductor cables.

Fig. 18: Protective conductor cables - PDC

Pos. 1 Two protective conductor cables to the sliding gantry

5.4 Gantry

5.4.1 Functional check of the fans in the generator electronic box

SI Function of E-box fans checked

Fig. 19: Generator electronic box

- Check whether the 4 fans of the generator electronic box are working.

5.4.2 Functional check of the HV transformer fans

SI Function of HV transformer fans checked

Check whether both fans located on the HV transformer are working.

5.4.3 Functional check of the C-box fans (only Sensation 10 / 16 / Cardiac)

SI Function of C-box fan checked

Check whether the C-box fan is working.

5.4.4 Functional check of the DMS fans

SI Function of DMS fans checked

Check whether both DMS fans are working.

5.4.5 Cleaning the slip ring compartment

PM Slip ring compartment and assembly cleaned

- Switch off the system.

⚠WARNING	**[hz_serdoc_F13G01U05M03]** **Dangerous voltages (up to 560 VDC) are present when the system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.** **Risk of electric shock!** ➪ **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).** ➪ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).** ➪ **Ensure via measurements that all voltages are switched off.** ➪ **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

- Remove the back gantry cover.

  Make sure to disconnect X350 and PE350 prior to removing the back cover completely!
- Remove the gantry funnel.

  Make sure to disconnect X312 A-D.
- Remove carbon dust from the
  - complete slip-ring compartment
  - slip ring assembly

  using a vacuum cleaner. **Do not use alcohol or other cleaning liquids**.

5.4.6 Cleaning/checking the power brush block

PM Power brush block cleaned

PM Power carbon brushes checked

NOTE	**Perform this section every 6 months.**

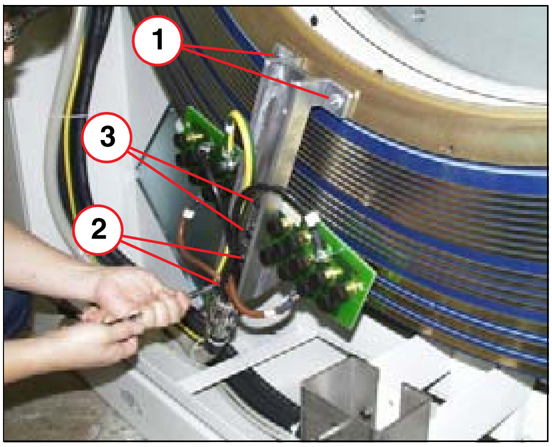


Fig. 20: Power brush assembly

Pos. 1	Support bracket: location of inner fastening screws
Pos. 2	Support bracket: location of outer fastening screws
Pos. 3	Brush block: location of mounting screws

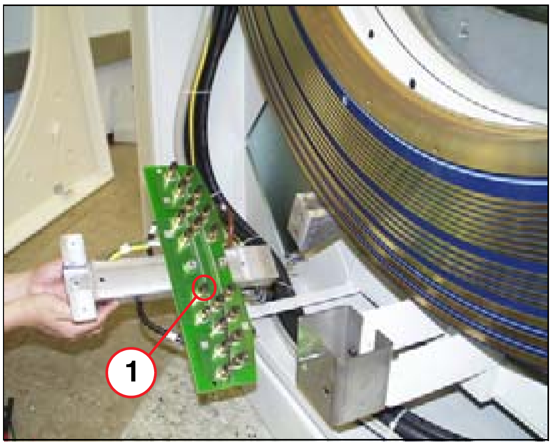


Fig. 21: Power brush assembly

Pos. 1 Check the each carbon brush (20 pieces)

- Remove the four mounting screws (two inner and two outer screws) of the power brush assembly and pry back the assembly. Be careful when handling the brush block; carbon brushes may become damaged!
- Remove the caps from the power carbon brushes and carefully pull out each brush. Put all brushes aside in a safe place.
- Use a vacuum cleaner to remove carbon dust from the power brush assembly. If necessary, you may use a clean soft-bristled toothbrush. Make sure that each brush guide is cleaned well to ensure functionality.
- Each carbon brush has to be **checked and replaced if**
  - any damage is visible (e.g. scratched or broken surface)
  - its length is out of tolerance. Each carbon brush has an engraved triangle which indicates the minimum tolerance.

NOTE

Ensure that the brush length will not go below the 12 mm limit prior to the next regular maintenance call.

- Reinstall all power brushes. Install all caps. Make sure all brushes move freely.
- Reinstall the power brush assembly. Refer to (Tab. 2 / p. 40)

Tab. 2 Information regarding screw torque

Screw	Tightening torque	Loctite 243
(1/Fig. 20 / p. 40)	3 Nm	yes
(2/Fig. 20 / p. 40)	24 Nm	no

5.4.7 Checking/replacing the voltage and control data brushes

PM Voltage and control data brushes checked

PM Control data brushes replaced

NOTE	Perform this section every 6 months.

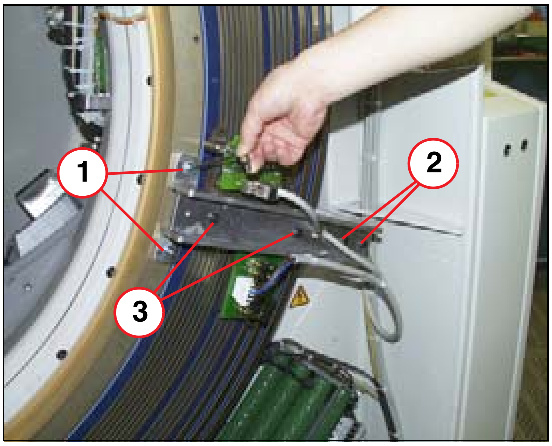


Fig. 22: Brush block for low power & control data

Pos. 1	Support bracket: 2 x inner fastening screws
Pos. 2	Support bracket: 2 x outer fastening screws
Pos. 3	Brush block: 2 x fastening screws

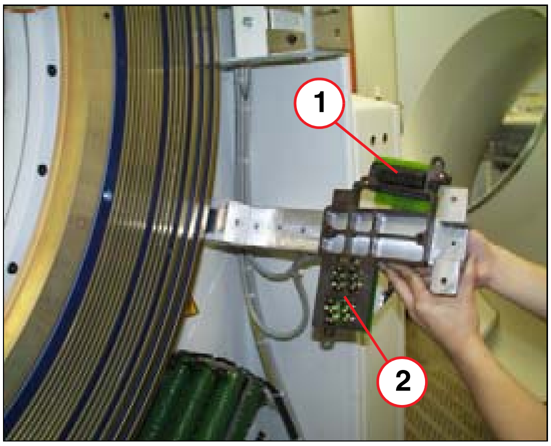


Fig. 23: Brush block for control data & low power

Pos. 1	Control data brushes
Pos. 2	Low power brushes

- Disconnect the X1 and X2 connectors.
- Loosen the two mounting screws of the power brush block (about 1- 2 turns).
- Remove the four mounting screws (two inner and two outer screws) and carefully pry back the assembly. Do not lose the plastic washers.
- Check the length of the brushes.

NOTE

Minimum length of carbon brushes required:

Control data brushes - 8mm

Low power brushes - 14mm

Ensure that the brush length will not fall below the minimum length before the next regular maintenance call.

- Replace the brushes if necessary.
- Reinstall the power brush assembly. Refer to (Tab. 3 / p. 42)
- Move the gantry by hand until the two adjustment holes of the slip ring are aligned with the brush block. An 8 mm pin may be helpful for this adjustment. Align the brush block and secure it with 2 screws.
- Connect the X1 and X2 connectors to the brush blocks.

Tab. 3 Information regarding screw torque

Screw	Tightening torque	Loctite 243
(1/Fig. 22 / p. 41)	3 Nm	yes
(2/Fig. 22 / p. 41)	28 Nm	no
(3/Fig. 22 / p. 41)	6 Nm +/- 10%	yes

5.4.8 Replacing the stator ring brush (only Sensation 10 / 16 / Cardiac)

PM Stator ring brush replaced

NOTE	**Perform this section every 6 months.**

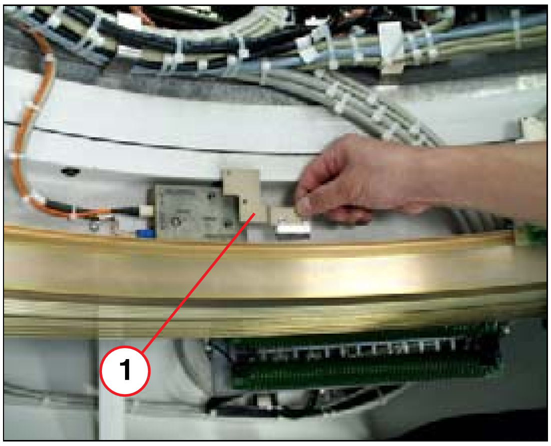


Fig. 24: Replacement of stator ring brush

Pos. 1 Stator ring brush

- Turn the rotating part of the gantry and position the transmitter of the data transmission line in the 6 o‘clock position.
- Remove the 2 mounting screws of the stator ring brush.
- Insert the new stator ring brush. Make sure that the brush has contact. As an indicator, the brush hairs should be slightly bent. Fasten the 2 screws.

5.4.9 Lubricating the rack and pinion gear of the UHR assembly (only Sensation 64, Cardiac 64)

NOTE	**No lubrication is required on the UHR-Plus. It is maintanance free.**

PM Rack and pinion gear lubricated (only Sensation 64, Cardiac 64, not UHR-Plus)

NOTE	**Perform this section every 12 months (full maintenance)**

- Clean old lubricant off the gear rack by using a lint-free cloth.

- Lubricate the gear rack with grease Isoflex Topas NCA52, appr. 0.4 ccm should be used.
- Evenly distribute the grease with a brush all over the gear rack.

NOTE	**Do not use the grease nipple for lubricating.**

5.4.10 Cleaning/checking the detector window

PM Detector window cleaned/checked

- Ensure that the radiation input window of the detectors is free from all contaminants such as contrast medium.
- Clean the detector window with alcohol, if necessary.
- Ensure that the radiation input window of the detector is not broken or severe scratched. This may impair the image quality of the system!
- Replace the radiation input window, if necessary.

5.4.11 Lubricating the main bearing

PM Main bearing lubricated

NOTE	**Perform this section every 12 months. (Full maintenance)**

NOTE	**Approximately 15 grams of lubricant should be used. Determine in advance how much lubricant is released with each application of the grease gun, so you will know how much lubricant to apply.**

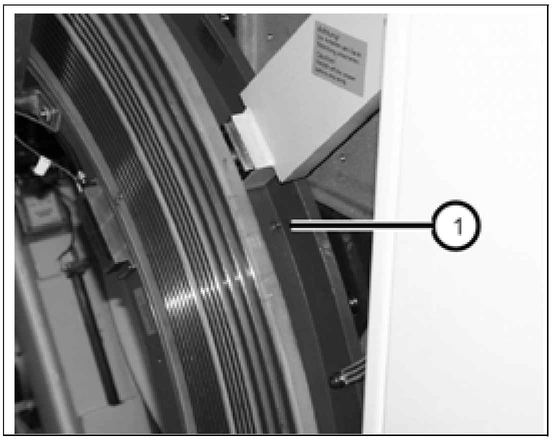


Fig. 25: Grease fitting, upper right side in the back of the gantry

- Connect the grease gun to the lubrication fitting (1/Fig. 25 / p. 43).
- Insert approximately 2.5 g of lubricant.

- Rotate the gantry by hand approximately 120 °.
- Insert another approximately 2.5 g of lubricant.
- Rotate the gantry by hand again approximately 120 °.
- Insert another approximately 2.5 g of lubricant.
- Rotate the gantry by hand several turns.
- Repeat the procedure.
- Disconnect the grease gun from the lubrication fitting.
- Check for and remove any excess lubricant at the bearing.
- Set service switch S301 to **ON.**

5.4.12 Replacing the filter in the dehumidifier

PM Filter in Dehumidifier replaced

Fig. 26: Dehumidifier: filter replacement

- Remove the filter assembly from the dehumidifier (Fig. 26 / p. 44).
- Replace the filter.
- Reinstall the filter assembly in the dehumidifier.

5.5 Power distribution cabinet

5.5.1 Replacing the air filter

PM Air filter in PDC replaced

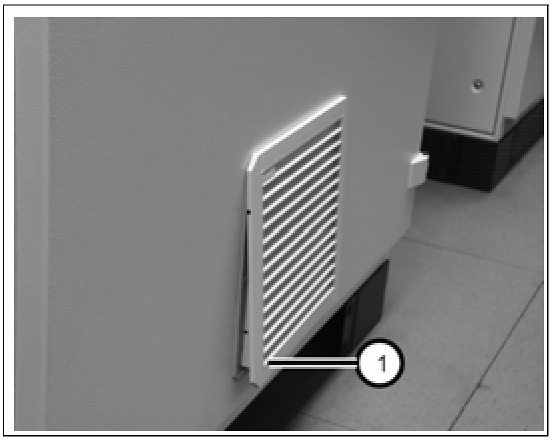


Fig. 27: PDC air filter

- Remove the air filter assembly from the front side of the PDC. (1/Fig. 27 / p. 45).
- Replace the filter.
- Reinstall the air filter assembly.

5.5.2 Checking/replacing the varistors and discharger

PM Varitors / discharger checked/replaced

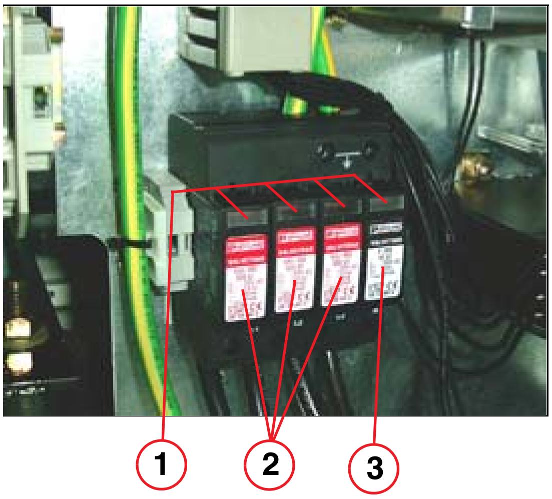


Fig. 28: Varistors and discharger

Pos. 1 Identification for defect

Pos. 2 Varistor

Pos. 3 Overvoltage protection

- A defective varistors / discharger will be displayed as “**defect**” in the small windows (pos. 1)
- If defective, replace it, as described in replacement of parts.

5.6 Water cooling system

5.6.1 Checking the water pressure

PM Water pressure checked

NOTE	**Switch off the water cooling system to measure the water pressure.**

- Place the manometer on the valve of the gantry cooling system.
- Open the front cover of the heat exchanger cabinet.

  ➪ **The nominal pressure for the cooling system is 2.0 bar +/- 0.2 bar.**
- Ensure the water pressure; if necessary, increase the water pressure by adding water. For detailed instructions, refer to service documentation “WCS- Replacement of parts”- CT02-023.841.05.xx.xx.
- Inspect the water pipes for leakage.

5.6.2 Cleaning the filter in the water-air cooling system

PM Filter cleaned

- Open the large tie wraps for the air hose at the air outlet (The hose closest to the rear of the cabinet, left or right side).
- Remove the hose from the outlet.
- Clean the filter using a vacuum cleaner.
- Connect the air hose and secure it using tie wraps.

  For detailed instructions, refer to service documentation “WCS- Replacement of parts”- CT02-023.841.06.xx.xx.

5.7 Imaging system

5.7.1 Cleaning the ICS, IRS and IES

PM Air inlets of ICS, and IES cleaned

PM Air inlets of IRS and air filters are cleaned

PM IRS air filter replaced

- Remove or open the front covers of the ICS, IRS and IES
- Clean the air inlets of the ICS, IRS and Wizard (IES) using a vacuum cleaner.

If the air filters on IRS1, IRS2 or IRS3 are damaged, exchange them:

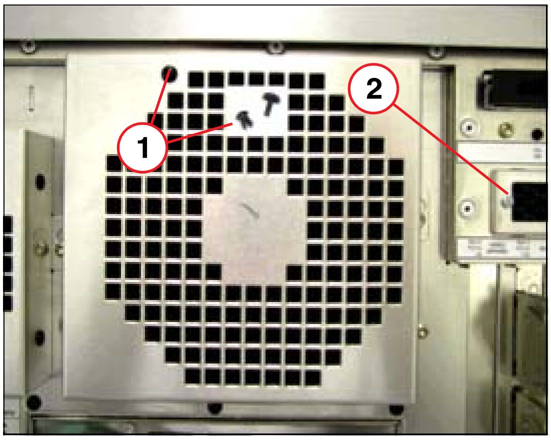


Fig. 29: Mounting of filter pads

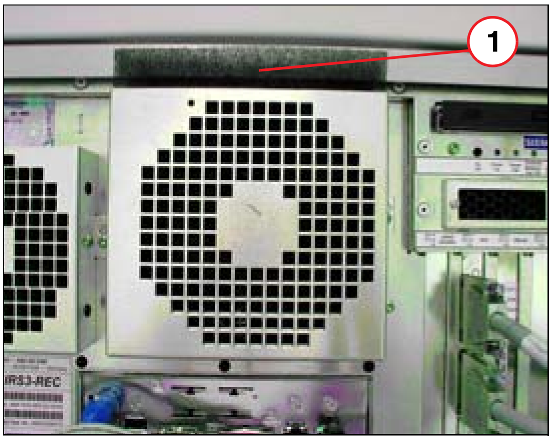


Fig. 30: Replacing the filter pad

- Remove the plastic, expanding rivet of the fan cover with a knife (1/Fig. 29 / p. 47). This rivet is only a transportation lock and cannot be installed again.
- Push the filter pads upward using a screwdriver (1/Fig. 30 / p. 47) and remove them from the filter frame.
- Insert the new filter into the filter frame and push it down
- Loosen the Phillips screw (2/Fig. 29 / p. 47) on the frame for the small filter pad (111mm x 21 mm).
- Move the frame to the left and remove it from the rack.
- Replace the filter pad.
- Mount the frame with the new filter pad onto the computer rack and attach it with the Phillips screw.

If the air filters at the IRSmx2s are damaged, exchange them:

- Open the front door using the key which is delivered with the computer.
- Open the six Torx screws TX10 and remove the cover and the air filter.

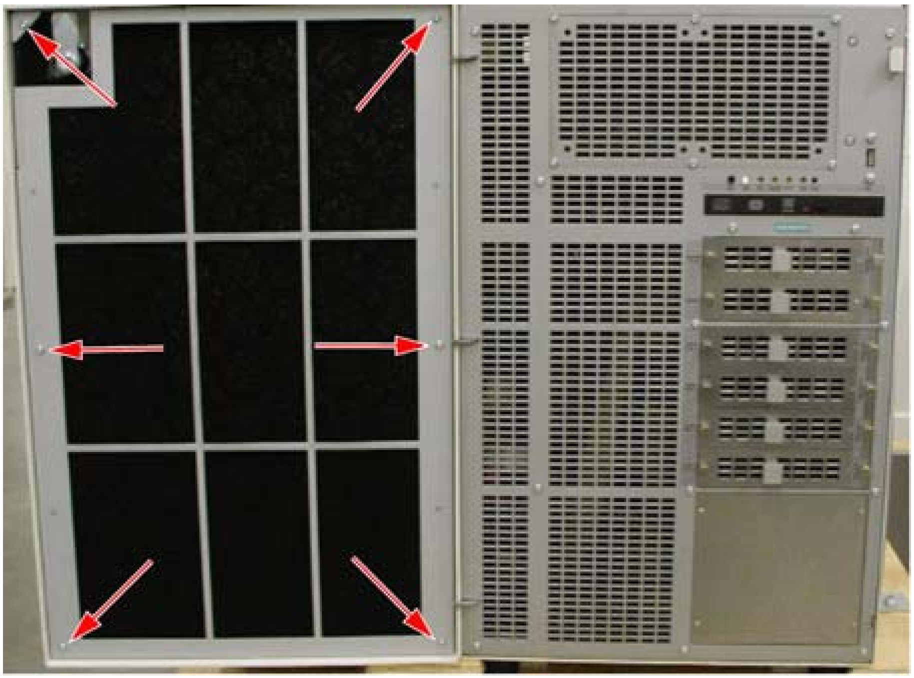


Fig. 31: Screws for air flow filter

- Insert the new air filter and secure it with the cover.

5.8 SRS Check

5.8.1 Work Steps for Som/7 VA11 and Som/5 VB30/VB38

5.8.1.1 Checking the status of the SRS installation (check only!)

PM Checking the status of the SRS installation

NOTE	The following steps ("Checking the status of the SRS installation" and "Performing System Management Configuration") have to be performed at the ICS (syngo CT Acquisition Workplace) and at the IES (syngo CT Workplace).

- In the **Local Service Home Menu** select **Configuration**
- Go to **SRS**
- Click > to start the SRS Function Check

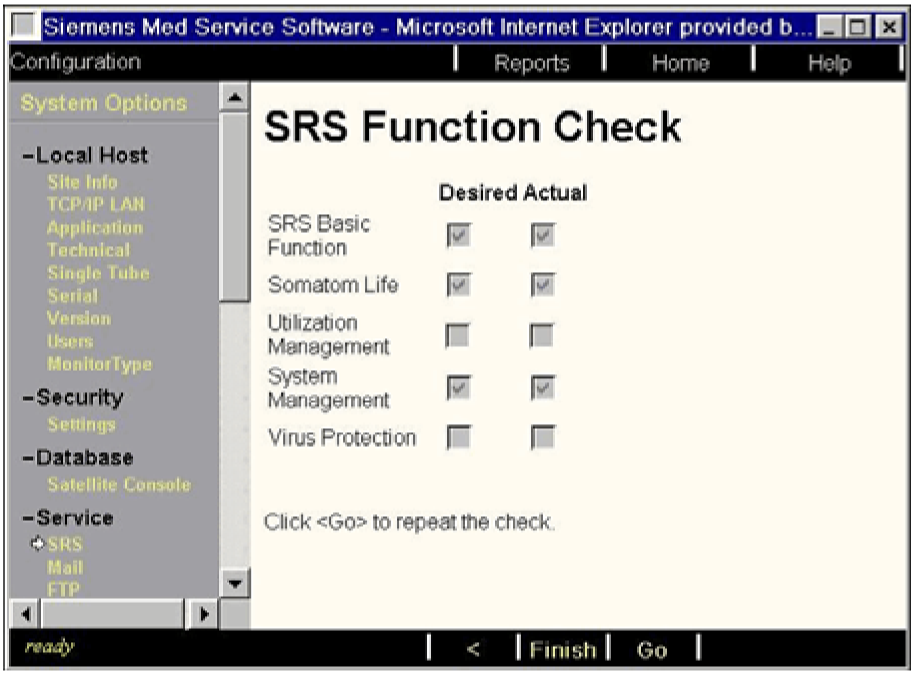


Fig. 32: SRS Function Check

- Under "Desired" at a minimum the following items must be enabled (checkbox is set, as shown in the figure above):
  - SRS Basic Function
  - Somatom Life
  - System Management

➪ If the “Actual” checkbox is set for all 3 items, SRS is configured correctly.
In this case, no further actions are necessary. You do not have to perform the work step “Performing System Management Configuration”.

➪ If the status “Actual” does not match “Desired”, an error message is output.
For further information on the error messages, a detailed description is added within the SRS configuration document, see (CT00-000.843.08 / Helpful hints).
SRS Basic Function must be completed successfully before you can continue with work step “Performing System Management Configuration”.

5.8.1.2 Checking the System Management Configuration

PM Checking the System Management Configuration

NOTE	**Please perform this step only if the SRS Basic Function from the SRS Funtion Check was completed successfully!**

- In the **Local Service Home Menu** select **Configuration**
- Select **System Management** -> **Agent Controls**

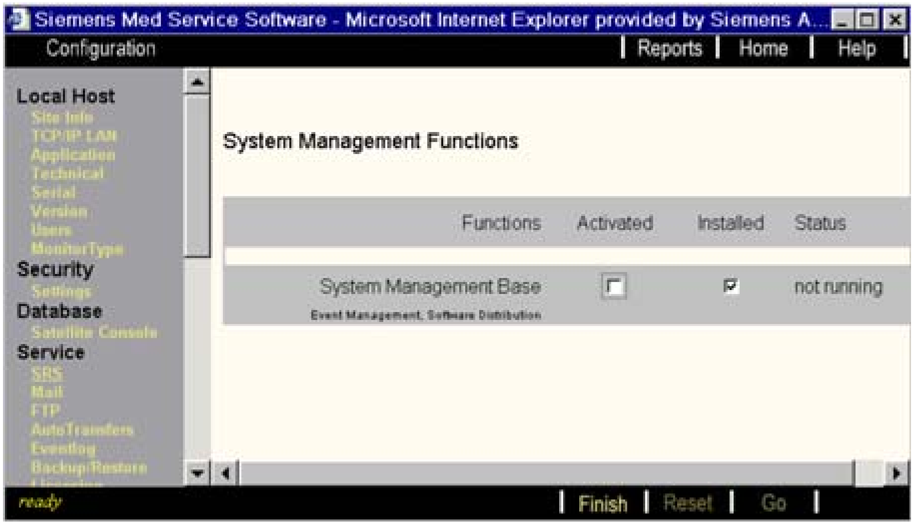


Fig. 33: System Management

NOTE	**Most likely the checkbox “Installed” is already enabled because the System Management Offline Installation is performed automatically during Somaris Installation.**

- Enable the checkbox “**Activated**” and press **GO** to complete the System Management configuration. During this configuration step, the system connects to the System Management server located in the SRS environment.

- If the check was successful, the following result will be shown:

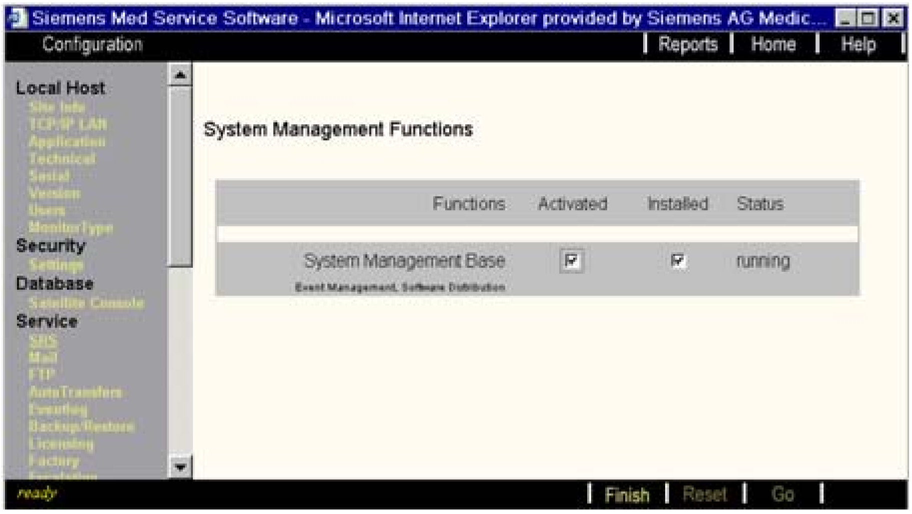


Fig. 34: Success

➪ Both checkboxes are enabled and the status is **"running"**.

➪ **NOTE**: Do not restart the system/application within the next 10 minutes because the template update is running in the background.

- If the check was not successful, an error result is output.

  See detailed error description, (CT00-000.843.08 / SRS Troubleshooting description)

5.8.2 Work Steps for Som/5 VB20/VB28

5.8.2.1 Checking the status of the SRS network connection (check only!)

PM Checking the status of the SRS network connection

NOTE	The following steps ("Checking the status of the SRS network connection" and "Performing System Management Configuration") have to be performed only at the ICS (syngo CT Acquisition Workplace).

- In the **Local Service Home Menu** select **Configuration**
- Go to **SRS**

- Click > to start the SRS Connection Check

Fig. 35: SRS Connection Check

➪ If the connection to the SRS server, e.g. 194.138.39.18 (DMZ Fürth for Europe), is **OK**, SRS is configured correctly.
In this case you can perform the work step "Performing System Management Configuration".

➪ If the connection to the SRS server is **NOT OK**, the CT is unable to dial out to the SRS server.
For further information on the error, a detailed description is added within the SRS configuration document, see (CT00-000.843.03 / Helpful hints).

5.8.2.2 Checking the System Management Configuration

PM Checking the System Management Configuration

NOTE	**Perform this step only if the SRS Connection Check was completed successfully.**

- In the **Local Service Home Menu** select **Configuration**

- Select **System Management** -> **Agent Controls**

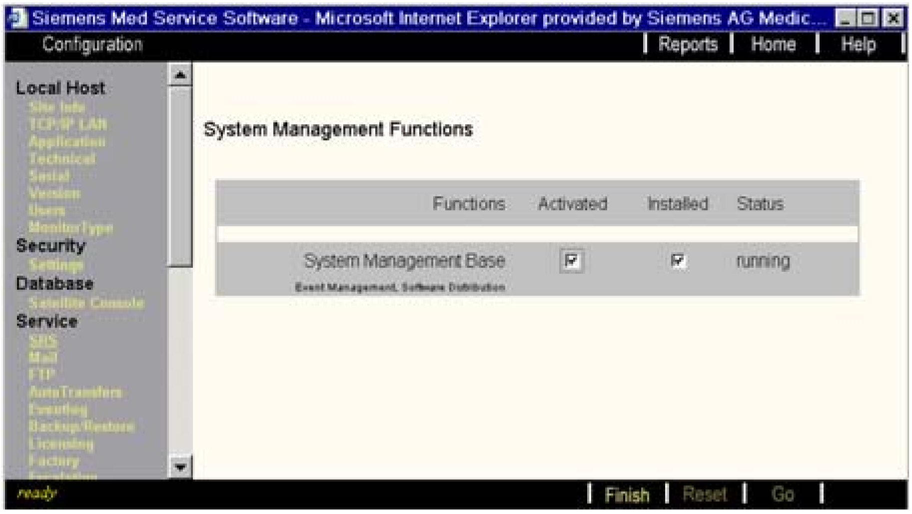


Fig. 36: Success

➪ If both checkboxes are activated and the status is "running", System Management is configured correctly. **In this case, you do not have to perform the following steps. The System Management check is finished.**

- If System Management is not configured correctly, the following appears:

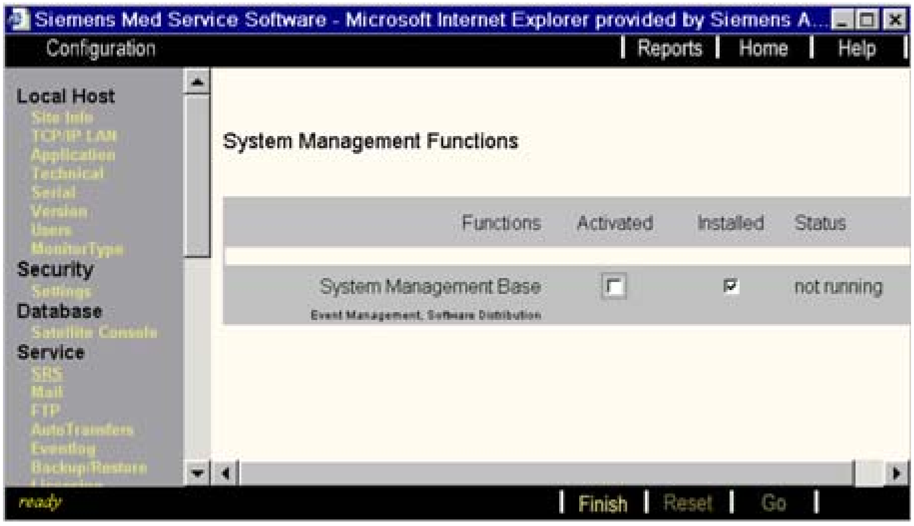


Fig. 37: System Management

NOTE	**Most likely the checkbox "Installed" is already enabled because the System Management Offline Installation is performed automatically during Somaris Installation.**

- Enable the checkbox "**Activated**" and press **GO** to complete the System Management configuration. During this configuration step, the system connects to the System Management server located in the SRS environment.
- If the check was not successful, an error result is output.
  See detailed error description, (CT00-000.843.03 / Helpful hints)

- If the check was successful, both checkboxes are activated (see picture from case 1, (Fig. 36 / p. 53))

  ➪ **NOTE**: Do not restart the system/application within the next 10 minutes because the template update is running in the background.

5.9 Next steps

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the 'Maintenance Protocol' provided. Depending on the maintenance schedule, you may either conclude the maintenance activities with the concluding test or proceed with further maintenance work according to the schedule (Recommended 'Image Quality Check' or, if necessary, first 'Preventive Parts Replacement').

5.9.1 Concluding test

Q Condition of the CT system checked

- Switch on the system and activate the service platform.
- Activate **Test Tools** --> **Rot Mode** and select the **slowest time**available.
- Press **<Go>** and rotate the gantry for approx. 1 minute to distribute the grease in the main bearing.
- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

6 Preventive parts replacement

6.1 General

⚠WARNING	[hz_serdoc_F13G01U12M01] **Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.** **Risk of accident and/or injury!** ➪ **The guidelines contained in this document have to be strictly followed.**

6.1.1 Preparations

NOTE	**All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the 'Maintenance Protocol' provided.**

6.1.2 Required tools and auxiliary materials

- Standard service kit

6.2 Uninterrupted power supply

6.2.1 Prerequisites

Tools and auxillary equipment

- Standard tool kit

Time and manpower

- 2 persons
- 1 hour

6.2.2 Replacing the standard UPS

PM **UPS replaced**

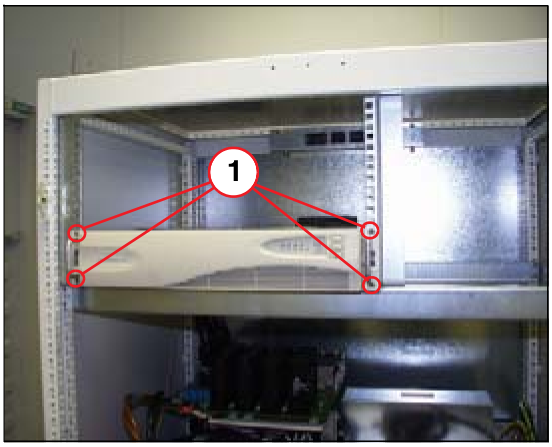
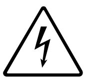


Fig. 38: UPS- uninterruptible power supply

Pos. 1 4x set screws



- The system must be switched off completely prior to any intervention of the system.

⚠WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when the system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- **Ensure via measurements that all voltages are switched off.**
- **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

- Check the status LED’s of the UPS. All LED’s must be off!
- Remove the four mounting screws (1/Fig. 38 / p. 57).
- Carefully pull the UPS halfway out of the PDC frame.
- Disconnect the plugs on the back side of the UPS
  - 1 male plug from the UPS connector "LOAD1"
  - 1 female connector of UPS-1 from the UPS connector "INPUT"
- Pull the UPS carefully out of the PDC frame.**Due to the heavy weight, a second person is required.**
- Connect the new UPS control cable (delivered with the UPS) to the “COMM” port at the UPS.
- Check the locations of the power plugs on the PDC. It will be difficult to find the location when the UPS is already halfway in the PDC.
- Insert the new UPS halfway into the PDC frame.
- Connect the male plug of UPS-1 to the UPS connector "LOAD1".
- Connect the female connector of UPS-1 to the UPS connector "INPUT".
- Route the UPS control cable from "COMM" port at the UPS to the PDC-X270 .
- Push the UPS completely into the PDC frame.
- Secure the UPS with four Allen screws(1/Fig. 38 / p. 57).
- Switch on the UPS

6.3 DMS cover Sensation 10 / 16 / Sensation Cardiac

6.3.1 Prerequisites

Tools and auxillary equipment

- Standard tool kit

Time and manpower

- 1 persons
- 1 hour

6.3.2 Check the correct attachment of the lead sheet metal.

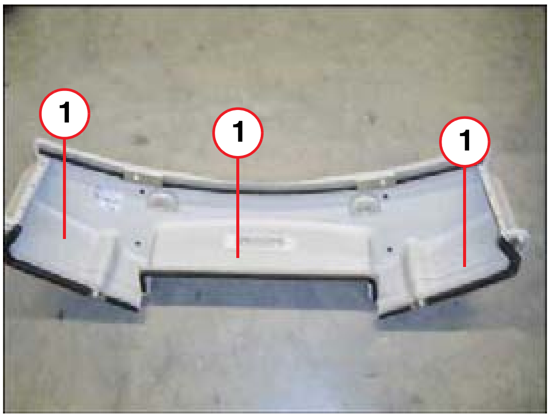


Fig. 39: DMS Cover
Pos. 1 Lead sheet metal

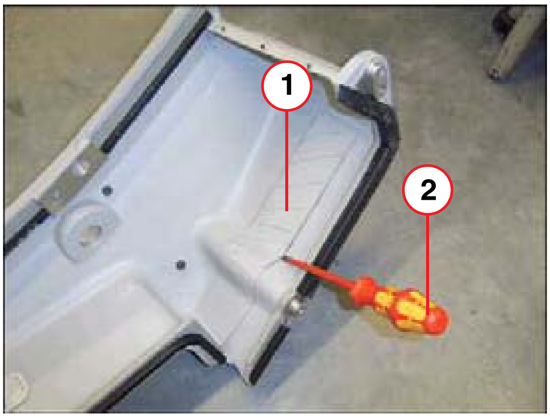


Fig. 40: DMS Cover
Pos. 1 Lead sheet metal
Pos. 2 Screwdriver

Check the correct attachment of the lead sheet metal.

- Check the lead sheet metal, that it is compact fixed with the cover.
- If a gap is visble anywhere along the lead sheet metal angle, use an screw driver to check the stability of the adhesive sealing. Insert the screw driver carefully in the gap and turn the screw driver carefully.
  - ➪ If the lead sheet metal is loose, replace the cover.

NOTE	**The cover must be checked until DMS serial number 51181767 resp. cover serial number 05112517** **The cover is free of charge for the corresponding serial numbers.** **Order number of cover: 10097152**

PM DMS cover replaced

6.4 Next steps

NOTE

The date of replacement as well as the date of next replacement has to be documented in the 'Maintenance Protocol' provided. Depending on the maintenance schedule, you may either conclude the maintenance activities with the concluding test or proceed with the maintenance according to the schedule (Recommended 'Image Quality Check').

6.4.1 Concluding test

Q Condition of the CT system checked

- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

7 Image quality - constancy

Q Image quality check completed

NOTE	The procedure for the image quality - constancy test is described in the “SOMATOM Operator Manual, chapter “Quality Assurance - Constancy Test”

8 Final steps

NOTE

You have completed the maintenance. All work performed on the system during maintenance and the results obtained have to be documented in the ’Maintenance Protocol’ provided. After performing the "Final Steps," provide the customer with the completed protocol and, depending on organizational requirements, return a copy of it to your regional unit or field service office.

8.1 Installing the unit covers

SIM **Covers and protective conductor cables are installed**

NOTE	All metal system covers are equipped with a plug or screw connection for the protective conductor. Ensure that the protective conductor cable is connected to this point when the system covers are installed. The protective conductor cable is included with the components.

8.2 Checking the protective system conductor

Note: Measure the protective conductor meter as described in the following product-specific contents.

- Make sure that the measuring instrument is calibrated according to manufacturer's regulations.
- Use a protective conductor meter to perform the test.

  Measuring instrument used: (type, manufac., serial no. and date of calibration)

- Unroll the test cable completely from the test device prior to performing the test to ensure correct results. Check the correct zero setting.
- **Enter the value measured into the corresponding protocol or checklist.**

8.3 General notes when measuring at the monitors

Always follow the note shown below when measuring at any monitor.

NOTE	**Disconnect the monitor signal cable to the computer. Other-wise, the graphics board will be destroyed.**

8.3.1 Measuring point 18” monitor

Fig. 41: 18 Inch Flachbildschirm
Pos. 1 Messpunkt für Schutzerdung

8.3.2 Measuring point 19” monitor

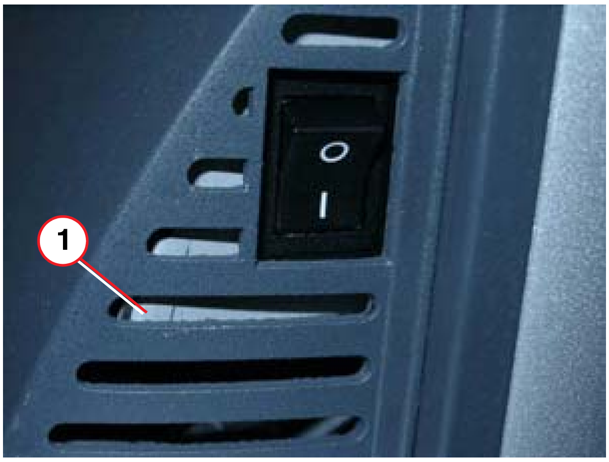


Fig. 42: 19 inch flat screen
Pos. 1 Measurement point

8.4 SOMATOM Sensation 10 / 16 / Cardiac / 40 / 64 / Cardiac 64 / Open

SIE **PDC Cabinet: Protective conductor resistance ≤100 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Protective ground terminal in the PDC

SIE **Gantry: Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE **Patient table (PHS): Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE **ICS Tower PC: Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE **IRS Cabinet: Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE **IES Tower PC: Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE **ICS Monitor: Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Measurement point of the respective monitor

SIE **IES Monitor: Protective conductor resistance ≤300 mΩ**

- From: Protective ground terminal in on-site power distribution
  To: Measurement point of the respective monitor

SIE **WCS Cabinet: Protective conductor resistance (mains connected directly) ≤100 mΩ**
Protective conductor resistance (connected via PDC) ≤300 mΩ

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE **UPS IES: Protective conductor resistance ≤100 mΩ**

- From: Power plug protective ground
  To: UPS Protective ground terminal

SIE **IES Tower PC: Protective conductor resistance ≤300 mΩ**

- From: Power plug protective ground
  To: Uncoated metallic part

8.5 CARE Vision CT option

SIE Overhead support: Protective conductor resistance ≤300 mΩ

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE Monitor trolley: Protective conductor resistance ≤300 mΩ

- From: Protective ground terminal in on-site power distribution
  To: Uncoated metallic part

SIE Monitor: Protective conductor resistance ≤300 mΩ

- From: Protective ground terminal in on-site power distribution
  To: Measurement point of the respective monitor

8.6 Cleaning the monitors

SIM **Monitors cleaned**

- Clean the monitors.
- Remove any dirt or contamination caused by maintenance work.

8.7 Testing the system application

Q Condition of the CT system checked

- To ensure that the system is in good working condition before turning it over to the customer, simulate several system functions typical of the customer.

9 Changes to previous Version

Chapter	Change	Reason
3	Reference to System Owner Manual for location of labels added.	Mod. No. 527346

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